Shaokang Jiang

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Education

University of California, Irvine — Irvine, CA
Ph.D. in Software Engineering, Advised by Daye Nam — *2025-2030 (expected)*

University of California San Diego — La Jolla, CA
M.S. in Computer Science and Engineering, Advised by Michael Coblenz — *2022-2025*
Thesis: Usability Analysis of Programming Assistants

University of Wisconsin-Madison — Madison, WI
B.S. in Computer Science, GPA: 4.00/4.00, graduated with distinction in the major — *Graduated 2021*

Publications

[1] S. Jiang and D. Nam, "Motif: Discovering and automating personal web workflows," *arXiv preprint arXiv:2607.10531*, Jul. 2026, Preprint.

[2] S. Jiang and D. Nam, "Beyond the prompt: An empirical study of cursor rules," in *23rd International Conference on Mining Software Repositories (MSR 2026)*, 27% acceptance rate, 2026.

[3] S. Jiang, A. Arya, S. Kweon, I. Liang, D. Kumar, and K. Vaccaro, "Long story short: Auditing U.S. political polarization in recommendations for long- vs. short-form videos on youtube," in *Proceedings of the ACM Web Conference (WWW 2026)*, 20% acceptance rate, 2026, pp. 6147–6158.

[4] J. A. Rayan, S. Jiang, N. Balaji, J. Wang, I. Gross, C. Biehle, and S. P. Dow, "Cueing the crowd: Llm-driven conversational cues across different meeting modalities increase topical diversity of generated ideas," in *Proceedings of the ACM Collective Intelligence Conference (CI 2025)*, 2025, pp. 297–316.

[5] S. Jiang, J. Koppel, and M. Coblenz, "Questions before answers? a comparison of codebase chatbots and tutorials for codebase learning," in *Plateau Workshop (Plateau 2025)*, 2025.

[6] S. Jiang and M. Coblenz, "An analysis of the costs and benefits of autocomplete in ides," in *Proceedings of the ACM on Software Engineering (FSE 2024)*, 25% acceptance rate, vol. 1, 2024, pp. 1284–1306.

Research Experience

Graduate Student Researcher — June 2026 – now
University of California, Irvine (supervised by Daye Nam and Emerson Murphy-Hill)

- Conducted a mixed-methods study of personal software built with AI coding agents, combining thematic analysis of 6,168 web and Reddit accounts with observation of eight people building tools for their own needs and a seven-day follow-up completed by six participants.
- Developed LLM-assisted analysis workflows for relevance filtering, qualitative coding, functionality evolution, and screen-recorded activity, with human validation; synthesized 17 themes describing creators' needs, development practices, and engineering challenges.
- Found that personal software requirements emerged through iteration: 68% of functionality arose during building. Creators' effort centered on verification, with 47% of building time spent testing or reading outputs and only 0.2% editing code.

Graduate Student Researcher — August 2025 – now
University of California, Irvine (supervised by Daye Nam)

- Analyzed human-AI collaboration using Cursor rules as a proxy, identifying four primary categories (Project, Convention, Guideline, LLM Directive) and revealing that developers adapt context types by programming language while applying traditional human-to-human documentation strategies to instruct AI. Published at **MSR 2026**.
- Designed and implemented a pipeline to collect and analyze Cursor rules from 401 GitHub repositories, leveraging prompted LLMs to scale up the qualitative coding process.
- Developed an ambient automation discovery system utilizing LLMs to passively observe web behavior and generate personalized programs, validated via an 8-person empirical study, revealing users inherently fail to recognize over 94% of their own automatable actions. Submitted to **CHI 2027**.

- Designed and implemented an asynchronous, natively cross-platform multimodal discovery agent (Rust, JavaScript, Chrome APIs) utilizing contextual memory and different LLM reasoning (e.g.,chain-of-thought, self-reflection, programming-as-reasoning, planning, and cross-verification) to mine web interactions into patterns. The system designs and generates scripts following human feedback, showing a 93% usefulness rating for all viewed patterns.

Graduate Student Researcher April 2024 – May 2025

University of California San Diego (supervised by Michael Coblenz)

- Found Copilot helps users focus on learning while increasing productivity; revealed developers prefer analyzing generated code over explanations, yet still rely on search engines for conceptual questions. Submitted to **EMSE**.
- Designed an eye-tracking experiment to evaluate the costs and benefits of Copilot for novice programmers, developing a universal eye-tracking platform for native VSCode to capture user interaction data across in-editor events.

Graduate Student Researcher April 2024 – May 2025

University of California San Diego (supervised by Kristen Vaccaro and Deepak Kumar)

- Audited YouTube's recommendation strategies for short and long-form content; discovered that long-form algorithms reinforce partisan echo chambers by aligning with seed videos, while short-form algorithms prioritize engagement by exposing users to more right-leaning content. Published at **WWW 2026**.
- Designed and implemented a pipeline of seeds collecting, matching, recommendation collecting, and political assessing, leveraging Selenium to harvest recommendations, git-diff for video matching, and prompt-engineered GPTs and BERT for political labeling.

Researcher April 2024 – December 2024

Independent (supervised by Jimmy Koppel and Michael Coblenz)

- Evaluated the impact of AI assistance on code comprehension among 15 experienced engineers, finding that LLM chatbots increased user confusion, while traditional tutorials improved high-level structural understanding but lacked low-level coding support. Published at **Plateau 2025**.

Graduate Student Researcher January 2023 – September 2023

University of California San Diego (supervised by Michael Coblenz)

- Found that autocomplete speeds up information acquisition but may not increase productivity with 32 programmers working with an unfamiliar API. Published and presented the paper at **FSE 2024**.
- Found that participants who used autocomplete learned more while spending less time reading documentation.
- Developed a web-based VSCode interaction and eye-tracking platform (JS/C#, Tobii Eye Tracker 5) to capture in-editor events, achieving a 90% latency reduction than previous solutions.

Technical Skills

- **Languages**: Java, Python, Rust, C, C++, C#, HTML/CSS/JavaScript, TypeScript, Matlab, LaTeX, GAMS, Markdown, Haskell, SQL, Embedded JavaScript, Blogger Template Language, Shell
- **Frameworks**: React, Node.js, Vue, Jest, Vitest, JSP, Flask, Expo (React Native), JUnit, Svelte, VitePress, Hexo, Electron, Chrome Extensions, VSCode Extensions, Puppeteer, Selenium, Tampermonkey, JavaFX, Servlets
- **Platforms/Tools**: Autogen, vllm, ollama, Git, Docker, GitHub, Google Cloud, Azure, Cloudflare (Workers and KV), Firebase, Oracle Cloud, VS Code, Visual Studio, IntelliJ, Eclipse, Nginx
- **Libraries/Services**: Auth0, Stripe, Matomo, MailJet, Daily.js, Socket.io, TensorFlow, WebGL, NLP.js, echarts, Matplotlib, echarts, PyQt, Pyramid, CherryPy, Three.js, and much more
- **Other Platforms**: Windows, Linux, MAC, JIRA, Trello, DevOps, Agile, JMP, Figma, Photoshop, Qualcoder, Taguette, Premiere, Audition, Blender, Excel, Word, PowerPoint

Awards and Activities

- Junior PC, MSR 2027 2027
- Student volunteer at FSE 2024 July 2024